

Demonstration of $T = \tan \beta l_t$

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1 System model

A shunt stub of normalized susceptance b , followed by a transmission line of characteristic impedance Z_0 and electrical length $\theta = \beta l_t$, followed by a second, identical shunt stub of susceptance b .

We define:

$$c = \cos \theta, \quad s = \sin \theta \quad (1)$$

2 Elementary ABCD matrices

Shunt stub (shunt admittance jbY_0):

$$M_{stub} = \begin{pmatrix} 1 & 0 \\ jbY_0 & 1 \end{pmatrix} \quad (2)$$

Transmission line of electrical length θ :

$$M_{line} = \begin{pmatrix} c & jZ_0s \\ jY_0s & c \end{pmatrix} \quad (3)$$

3 ABCD matrix of the full system

$$M = M_{stub} \cdot M_{line} \cdot M_{stub} \quad (4)$$

Step 1:

$$M_{stub} \cdot M_{line} = \begin{pmatrix} 1 & 0 \\ jbY_0 & 1 \end{pmatrix} \begin{pmatrix} c & jZ_0s \\ jY_0s & c \end{pmatrix} = \begin{pmatrix} c & jZ_0s \\ jY_0(bc + s) & c - bs \end{pmatrix} \quad (5)$$

($Y_0Z_0 = 1$)

Step 2:

$$M = \begin{pmatrix} c & jZ_0s \\ jY_0(bc + s) & c - bs \end{pmatrix} \begin{pmatrix} 1 & 0 \\ jbY_0 & 1 \end{pmatrix} \quad (6)$$

$$A = c - bs, \quad B = jZ_0s \quad (7)$$

$$C = jY_0(bc + s) + (c - bs)(jbY_0) = jY_0 [2bc + (1 - b^2)s], \quad D = c - bs \quad (8)$$

$$M = \begin{pmatrix} c - bs & jZ_0s \\ jY_0 [2bc + (1 - b^2)s] & c - bs \end{pmatrix} \quad (9)$$

4 Input reflection coefficient

For an ABCD two-port terminated in Z_0 :

$$\rho = \frac{A + B/Z_0 - CZ_0 - D}{A + B/Z_0 + CZ_0 + D} \quad \left(\rho = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} \right) \quad (10)$$

with $Z_{in} = \frac{V_1}{I_1} = \frac{AZ_0 + B}{CZ_0 + D}$

Since $A = D$:

$$\rho = \frac{B/Z_0 - CZ_0}{2A + B/Z_0 + CZ_0} \quad (11)$$

Numerator:

$$\frac{B}{Z_0} = js, \quad CZ_0 = j[2bc + (1 - b^2)s] \quad (12)$$

$$\frac{B}{Z_0} - CZ_0 = j[s - 2bc - (1 - b^2)s] = j[b^2s - 2bc] = ju, \quad u \equiv b(bs - 2c) \quad (13)$$

Denominator:

$$2A + \frac{B}{Z_0} + CZ_0 = 2(c - bs) + j[2bc + (2 - b^2)s] \quad (14)$$

Hence:

$$\rho = \frac{ju}{2(c - bs) + j[2bc + (2 - b^2)s]} \quad (15)$$

5 Zeros of reflection

$$\rho = 0 \iff u = 0 \iff b(bs - 2c) = 0 \quad (16)$$

$$b = 0 \quad \text{or} \quad b = \frac{2c}{s} = 2 \cot \theta \quad (17)$$

6 Location of the maximum between the two zeros

We look for the extremum of $u(b) = b^2s - 2bc$:

$$\frac{du}{db} = 2bs - 2c = 0 \implies b_{max} = \frac{c}{s} = \cot \theta \quad (18)$$

This point is the midpoint between $b = 0$ and $b = 2 \cot \theta$ (the two zeros).

7 $A = 0$ at the maximum

$$A_{max} = c - b_{max}s = c - \frac{c}{s} \cdot s = 0 \quad (19)$$

8 Computation of ρ_{max}

Numerator, at $b = b_{max} = c/s$:

$$u_{max} = b_{max}^2 s - 2b_{max}c = \frac{c^2}{s} - \frac{2c^2}{s} = -\frac{c^2}{s} \quad (20)$$

Denominator, using $2(c - b_{max}s) = 0$ cause $A_{max} = 0$:

$$2bc + (2 - b^2)s \Big|_{b=c/s} = \frac{2c^2}{s} + \left(2 - \frac{c^2}{s^2}\right)s = \frac{2c^2}{s} + 2s - \frac{c^2}{s} = \frac{c^2 + 2s^2}{s} \quad (21)$$

Ratio of moduli:

$$\rho_{max} = \frac{\frac{|u_{max}|}{\left|\frac{c^2 + 2s^2}{s}\right|}}{\frac{c^2/s}{(c^2 + 2s^2)/s}} = \frac{c^2}{c^2 + 2s^2} \quad (22)$$

$$\boxed{\rho_{max} = \frac{c^2}{c^2 + 2s^2}} \quad (23)$$

9 Isolating $T = \tan \theta$

Dividing numerator and denominator by c^2 :

$$\rho_{max} = \frac{1}{1 + 2 \tan^2 \theta} = \frac{1}{1 + 2T^2} \quad (24)$$

$$1 + 2T^2 = \frac{1}{\rho_{max}} \implies T^2 = \frac{1}{2} \left(\frac{1}{\rho_{max}} - 1 \right) \quad (25)$$

$$\boxed{T = \tan \beta l_t = \sqrt{\frac{1}{2} \left(\frac{1}{\rho_{max}} - 1 \right)}} \quad (26)$$